

Technical Challenge – Bioinformatician Position (VHIO 2026-056-01)

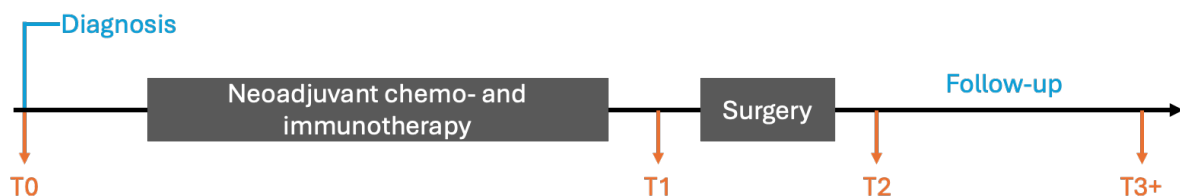
Project scenario

Imagine a clinical trial involving 200 patients with ovarian cancer who are receiving neoadjuvant immunotherapy prior to surgery.

The overall aim of the trial is to understand and predict response to immunotherapy and to investigate whether molecular changes detected in the blood can provide information about treatment response and subsequent disease outcome.

Plasma samples are collected at several time points:

- T0 – Diagnosis / pre-treatment: before initiation of immunotherapy
- T1 – Post-treatment / pre-surgery: following neoadjuvant immunotherapy and immediately before surgery
- T2 – Post-surgery: following surgical resection
- T3+ – Follow-up: subsequent longitudinal samples collected during clinical follow-up



Plasma samples are analysed using Oxford Nanopore sequencing.

Additional clinical (and for a subset of patients also molecular information) is available, including treatment response, pathological findings from surgery, and clinical follow-up data.

Research aims/questions:

The project has several related aims:

- Detection of circulating tumour DNA (ctDNA) in plasma.
- Detection of molecular residual disease (MRD) following treatment and surgery.
- Exploring whether information obtained from plasma samples can be used to predict treatment response and/or patient outcome.

Task/question:

You have been asked to help develop the computational strategy for analysing these data.

How would you approach the analysis of the sequencing data with the above outlined research questions in mind?

(You may outline a potential computational workflow from the raw sequencing data through to identifying a tumour-associated signal in plasma. You may discuss relevant tools, methods, features or analytical approaches that you would consider. You do not need to provide code or a detailed implementation.)

Preparation

Please prepare your thoughts in advance and be ready to discuss/explain your approach (10-15mins) during the interview.

You may use a small number of slides, diagrams, or other material to illustrate your ideas if you find this helpful, but this is not required.

No coding is expected.

There is no single correct answer. The main aim of this exercise is just to guide the interview discussions and to understand how you think and how you would approach a realistic bioinformatic challenge relevant to the type of work carried out in the group.

You do not need to have previous experience specifically with cfDNA, cancer genomics, immunotherapy, or Oxford Nanopore sequencing to answer the challenge. We are interested in your reasoning and approach to the problem.